Linear Algebra Basics

PART A — VECTORS

1. Which of the following represents a vector?

A. 25
B. [2, 5, 7]
C. 2 + 5 + 7
D. x = 10

2. If x = [2, 3] and y = [4, 5], what is x + y?

A. [6, 8]
B. [8, 15]
C. [2, 2]
D. [4, 3]

3. If x = [2, 3], what is 4x?

A. [6, 7]
B. [8, 12]
C. [2, 12]
D. [4, 3]

4. What does the magnitude of a vector represent?

A. Its direction
B. Its length
C. Its number of elements
D. Its number of dimensions only

5. What is the Euclidean norm of the vector [3, 4]?

A. 3
B. 4
C. 5
D. 7

6. Two vectors are orthogonal when:

A. They have the same magnitude
B. Their dot product is zero
C. Their values are identical
D. Their magnitudes are zero

7. What is the dot product of [1, 2] and [3, 4]?

A. 5
B. 7
C. 10
D. 11

8. If two non-zero vectors have a dot product of zero, they are:

A. Parallel
B. Orthogonal

C. Identical
D. Linearly dependent

PART B — MATRICES

9. A matrix with 4 rows and 3 columns has dimensions:

A. 4 × 3
B. 3 × 4
C. 12 × 1
D. 4 × 4

10. Which of the following is a square matrix?

A. 2 × 3
B. 3 × 2
C. 4 × 4
D. 5 × 3

11. What does transposing a matrix do?

A. Multiplies all elements by -1
B. Interchanges rows and columns
C. Converts all elements to zero
D. Divides every element by its mean

12. If matrix A has dimensions 3 × 5, what are the dimensions of A^T?

A. 3 × 5
B. 5 × 3
C. 3 × 3
D. 5 × 5

13. Which pair of matrices can be added?

A. 2 × 3 and 3 × 2
B. 2 × 3 and 2 × 3
C. 3 × 4 and 4 × 3
D. 2 × 2 and 3 × 3

14. Matrix multiplication AB is possible when:

A. Rows of A = rows of B
B. Columns of A = columns of B
C. Columns of A = rows of B
D. Both matrices are square

15. If A is 3 × 4 and B is 4 × 2, what is the dimension of AB?

A. 3 × 2
B. 4 × 4
C. 2 × 3
D. 3 × 4

16. If A is 2 × 3 and B is 3 × 4, what is the dimension of AB?

A. 2 × 4
B. 3 × 3

C. 4 × 2
D. 2 × 3

17. Which statement about matrix multiplication is generally true?

A. AB = BA always
B. AB and BA are always defined
C. AB is not necessarily equal to BA
D. Matrix multiplication is only possible for square matrices

PART C — LINEAR COMBINATIONS

18. A linear combination of vectors involves:

A. Only multiplication
B. Only addition
C. Scaling vectors and adding them together
D. Multiplying matrices only

19. Suppose:

v1 = [1, 0]

v2 = [0, 1]

Which vector can be represented as a linear combination of v1 and v2?

A. [2, 3]
B. [1, 1]
C. [5, -2]
D. All of the above

20. If:

v2 = 3v1

then v1 and v2 are:

A. Orthogonal
B. Linearly dependent
C. Linearly independent
D. Perpendicular

21. A set of vectors is linearly independent if:

A. Every vector has the same magnitude
B. No vector can be represented as a combination of the others
C. All vectors are perpendicular
D. All vectors contain positive values

22. If one feature can be exactly calculated from another feature, the two features are:

A. Orthogonal
B. Independent
C. Redundant
D. Random

PART D — RANK

23. The rank of a matrix represents:

A. Total number of elements
B. Number of rows
C. Number of linearly independent rows or columns
D. Number of diagonal elements

24. A matrix whose columns are all multiples of one another will have:

A. Rank 0
B. Rank 1
C. Rank equal to the number of columns
D. Infinite rank

25. If a dataset has 10 features but some features are completely redundant, the effective number of independent directions may be:

A. Greater than 10
B. Exactly 10
C. Less than 10
D. Infinite

PART E — GEOMETRIC INTERPRETATION

26. A vector can be interpreted geometrically as:

A. A point only
B. A direction and magnitude
C. A scalar value only
D. A matrix

27. What happens when a vector is multiplied by a scalar greater than 1?

A. Its direction always changes
B. Its magnitude increases
C. It becomes orthogonal
D. It becomes zero

28. Multiplying a vector by -1:

A. Changes its magnitude to zero
B. Reverses its direction
C. Makes it orthogonal to every vector
D. Removes its dimensions

29. The angle between two vectors can be determined using:

A. Their dot product and magnitudes
B. Their number of elements only
C. Their matrix dimensions
D. Their rank only

30. If two vectors point in exactly the same direction, the angle between them is:

A. 0°
B. 45°
C. 90°
D. 180°

PART F — PROJECTION

31. The projection of one vector onto another can be understood as:

A. The component of one vector along the direction of another
B. The difference between two vectors
C. The magnitude of both vectors multiplied together
D. A completely new random vector

32. If vector a is projected onto vector b, the projection tells us:

A. How much of a lies in the direction of b
B. How many elements are in b
C. The rank of a
D. The determinant of b

33. If two vectors are orthogonal, the projection of one onto the other is:

A. Maximum
B. Zero
C. Equal to its magnitude
D. Always one

34. Why is projection useful when representing data?

A. It can represent information using a chosen direction
B. It always increases dimensionality
C. It removes all information
D. It changes categorical data into numerical data

PART G — EIGENVALUES AND EIGENVECTORS

35. An eigenvector of a matrix is a vector whose:

A. Direction remains unchanged after the transformation, apart from scaling
B. Magnitude always becomes zero
C. Values always become positive
D. Number of elements changes

36. In the equation:

$A v = \lambda v$

what is v?

A. Matrix
B. Eigenvalue
C. Eigenvector
D. Determinant

37. In the equation:

$A v = \lambda v$

what is λ?

A. Eigenvector
B. Eigenvalue
C. Matrix rank
D. Matrix inverse

38. If an eigenvalue is 5, what does this generally indicate about its associated eigenvector?

A. The vector is rotated by exactly 5°
B. The vector is scaled by a factor of 5
C. The vector becomes zero
D. The vector becomes orthogonal

39. If an eigenvalue is zero, then:

A. The associated transformed vector is zero
B. The eigenvector does not exist
C. The matrix must be identity
D. The vector becomes larger

40. Eigenvectors are useful for identifying:

A. Important directions associated with a matrix transformation
B. Missing values in a dataset
C. Number of observations
D. Categorical variables

PART H — COVARIANCE AND VARIANCE

41. Variance measures:

A. The average value of a variable
B. How much a variable varies around its mean
C. The number of observations
D. The relationship between two different variables

42. If all values in a variable are exactly the same, its variance is:

A. 0
B. 1
C. -1
D. Infinite

43. Covariance measures:

A. How two variables vary together
B. The mean of one variable
C. The number of features
D. The magnitude of a vector

44. A positive covariance generally indicates:

A. Both variables tend to increase together
B. One variable always decreases
C. There is no relationship
D. Both variables are constant

45. A negative covariance generally indicates:

A. Both variables increase together
B. One variable tends to increase when the other decreases
C. Both variables are identical
D. Both variables have zero variance

46. If two variables have covariance close to zero, it indicates:

A. They definitely have no relationship of any kind
B. There is little or no linear relationship detected
C. They must be identical
D. They must be orthogonal vectors

PART I — APPLIED LINEAR ALGEBRA

47. Consider a dataset represented as a matrix where rows represent observations and columns represent features. What does each row represent?

A. A feature
B. An observation/sample
C. A matrix transformation
D. An eigenvector

48. In a dataset matrix, each column generally represents:

A. An observation
B. A feature/variable
C. An eigenvalue
D. A class boundary

49. Suppose two features contain almost exactly the same information. What is the most likely consequence?

A. Increased redundancy
B. Increased independence
C. Guaranteed increase in useful information
D. Orthogonality

50. A dataset contains 100 features, but many of them are strongly correlated. What might this suggest mathematically?

A. The data may have fewer effective independent directions than 100
B. The data must have exactly 100 independent directions
C. The data cannot be represented as a matrix
D. All features are orthogonal

PART J — THINK BEFORE YOU ANSWER

51. A transformation changes a vector from [2, 1] to [6, 3]. What can you immediately conclude?

A. The direction has changed
B. The vector has been scaled by 3
C. The vector has become orthogonal
D. The vector has been translated

52. Suppose three vectors lie along the same line. Their maximum number of linearly independent directions is:

A. 0
B. 1
C. 2
D. 3

53. If a matrix has highly dependent columns, its rank will generally be:

A. Higher
B. Lower
C. Infinite
D. Unaffected in all cases

54. A vector has a large magnitude but points in the same direction as another vector. What can be said about the two vectors?

A. They must be orthogonal
B. They may be scalar multiples of each other
C. They must be independent
D. They must have zero dot product

55. Which mathematical concept is most directly related to finding important directions in a linear transformation?

A. Eigenvectors
B. Mean
C. Median
D. Mode